

SECTION 1 — MODELS

[Chat Model Output]

LangChain is a development framework designed to build applications that leverage large language models (LLMs) by connecting them to external data sources and tools, enabling more complex and context-aware capabilities.

[Embedding Output] Vector length: 3072, First 3 values: [-0.01682972, 0.01739382, 0.023140239]

SECTION 2 — PROMPTS

[ChatPromptTemplate Output]

Virat Kohli

[MessagesPlaceholder Output]

Yes, I do! Your name is Rahul.

SECTION 3 — STRUCTURED OUTPUT

[Structured Output]

Title: Interstellar

Sentiment: positive

Rating: 9/10

Pros: ['visual masterpiece', 'stunning practical effects and cinematography', 'incredible emotional depth of the story', 'legendary Hans Zimmer's score']

Cons: ['third act gets a bit confusing and hard to follow']

Summary: A visually stunning and emotionally deep film with a legendary score, though the third act can be confusing.

SECTION 4 — OUTPUT PARSERS

[PydanticOutputParser Output]

Name: Arjun Singh, Age: 32, City: Mumbai

SECTION 5 — CHAINS + RUNNABLES

[Sequential Chain Output — Poem translated to Hindi]

यहाँ कविता का हिन्दी अनुवाद है:

आकाश रोता है, ढोलक-सी थाप का गान,
धरती पीती है जी भरकर, हरियाली का उत्सव महान,
ठंडी मॉनसून की ताज़गी भरी बौछार।

[Parallel + Passthrough + Lambda Output]

Joke: Why do programmers prefer dark mode?

Because light attracts bugs!

Word count: 10

SECTION 6 — DOCUMENT LOADERS + TEXT SPLITTERS

[Document Loader + Text Splitter]

Original docs: 5

After splitting: 7 chunks

First chunk: LangChain is a framework designed to simplify the creation of applications using large language models. It provides tools for chaining components together in powerful ways.

First chunk metadata: {'source': 'intro.txt'}

SECTION 7 — VECTOR STORE + RETRIEVER + RAG

[Similarity Search for 'What is RAG?']

Result 1: Retrieval-Augmented Generation (RAG) is a technique that combines information retrieval with text generation. It first retrieves relevant documents from a knowledge base, then uses them as context for

Result 2: Agents in LangChain use tools to take actions in the world. An agent decides which tool to call, runs it, and uses the result to continue reasoning until it reaches a final answer.

[RAG Chain Output]

Agents in LangChain use tools to take actions in the world. An agent decides which tool to call, runs it, and uses the result to continue reasoning until it reaches a final answer.

SECTION 8 — TOOLS + TOOL CALLING

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[Tool Properties]

add.name: add

add.description: Add two numbers together and return the result

add.args: {'a': {'title': 'A', 'type': 'integer'}, 'b': {'title': 'B', 'type': 'integer'}}

[Tool Calls Requested by Model]

[{'name': 'multiply', 'args': {'b': 7, 'a': 15}, 'id': '821f19b8-7de0-4e21-9776-3ac4a0526e66', 'type': 'tool_call'}]

[Tool 'multiply' Result]: 105

[Final Answer from Model]

15 multiplied by 7 is 105.

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REVISION COMPLETE — All concepts covered!

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